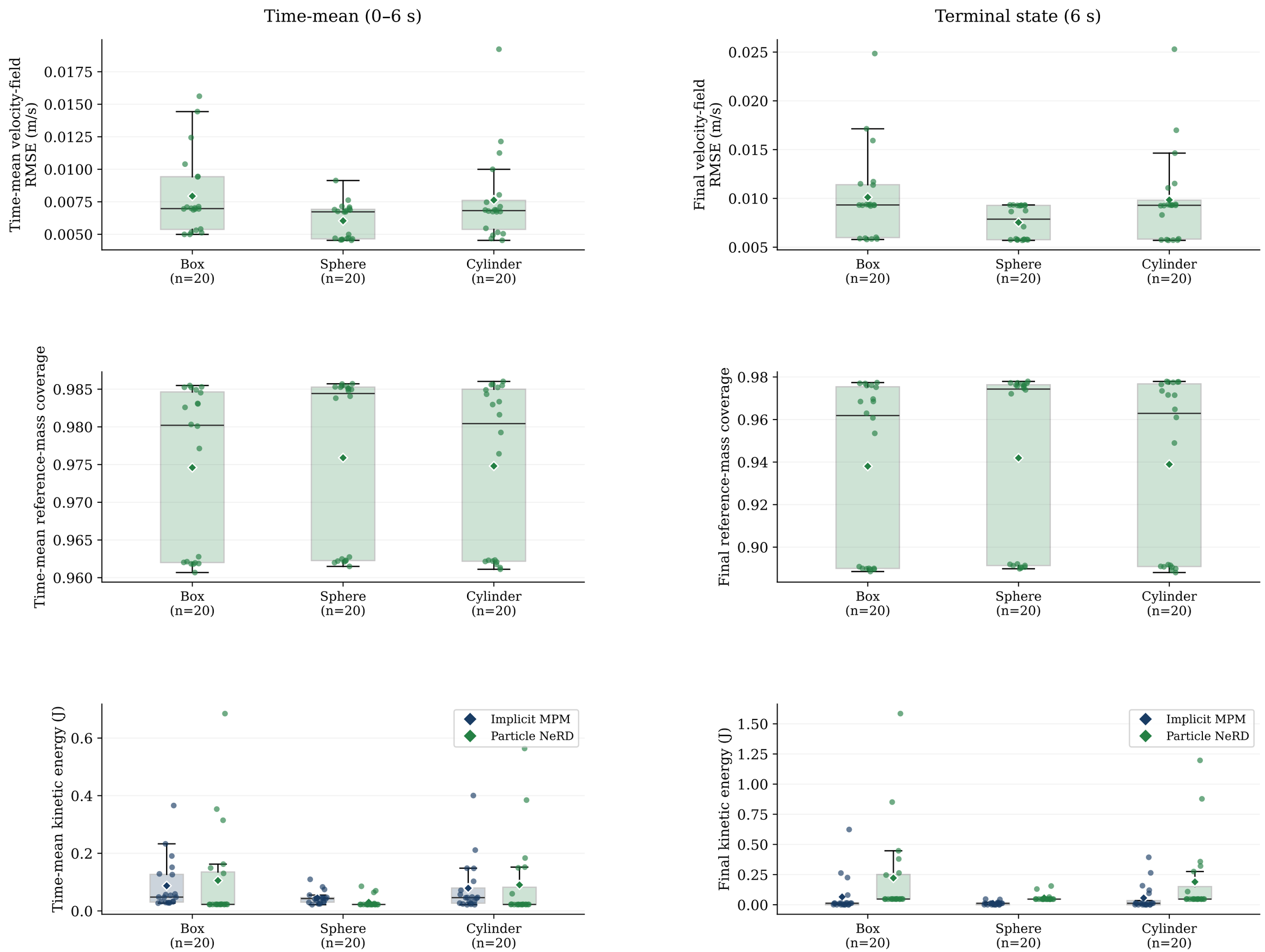

Figure 4: Frozen object-model checkpoint 9,000. Rows show velocity-field RMSE, the fraction of reference mass covered by shared occupied cells, and particle kinetic energy. Left column: time-mean over 0–6 s; right column: terminal state at 6 s. Velocities use matched consecutive-position differences. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.